MATHEMATICS CURRICULUM

THEME: NUMBERS AND NUMERATION **JSS ONE**

SUB-THEME: WHOLE NUMBERS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Whole Numbers	Students should be able to: 1. count and write in millions and billions 2. count and write in trillions 3. apply the counting, writing, and reading of large numbers in everyday life 4. solve problems in quantitative aptitude reasoning using large numbers	Counting in: i. Millions and billions ii. Trillions ii. Quantitative reasoning	Leads students to: 1. writes and read in millions and billions 2. uses counting charts to count in trillions 3. guides students in counting, writing and reading in large quantities 4. leads students to solve some problems on quantitative aptitude .	1. Count, write and read from 99,990 to 1,000,000 in tens and units. 2. Count, write, and read from 1,000, 000 to 100,000,000 in hundreds and thousands. 3. Count, write and read in trillions 4. Count, write and read in large quantities.	1. Charts of numbers in millions and billions, flash cards, etc 2. Charts of numbers in millions and billions, newspapers, 3. Flash cards, etc	Students to: 1. count, write and read in millions and billions 2. count, write and read in trillions 3. Solve problems on quantitative aptitude using large numbers.
2. LCM	Students should be able to: 1. identify common multiples of two or more whole numbers. 2. find the LCM of whole numbers.	LCM of whole numbers	Leads students to: 1. identify common multiples of two or more whole numbers 2. solve problems involving LCM by: (a) factor method (b) multiple method (c) Index method.	1. Identify common multiples of given numbers. 2. Solve problems involving LCM of two or three whole numbers using the three different methods	Number charts.	Students to: 1. Find the common multiples of given whole numbers. 2. Find the LCM of given whole numbers

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			TEACHER	STUDENTS		
3. HCF	Students should be able to: 1. identify common factors of whole number; 2. find the HCF of whole numbers; 3. identify the difference between LCM and HCF; 4. solve problems on quantitative aptitude involving LCM and HCF whole numbers.	i. HCF of whole numbers. ii. LCM and HCF of given whole numbers. iii. Quantitative reasoning.	1. Leads students to: i. identify common factors of two or more whole numbers using common factor and prime factor methods. ii. find the H C F of some whole numbers, eg HCF of 15 and 80 . iii. identify the difference between LCM and HCF. 2. Guides students in prime factorization to express a given number as the product of its prime factors as follows: 28 = 2 x 14 = 2 x 2 x 7 :. The prime factors of 28 are 2 and 7. 3. Leads students to solve problems on quantitative aptitude e.g. find the missing number: 28 $\longrightarrow$ 2^2. 7 84 $\longrightarrow$?	1. Write common factors and highest common factors of a given set of numbers. 2. Express whole numbers as prime factors. 3. Solve some given problems to find the HCF of whole numbers. 4. Identify the difference between LCM and HCF. 5. Solve related problems on quantitative aptitude.	HCF Chart containing worked examples. HCF/LCM Charts	students to: 1. find the HCF of given whole numbers; 2. express given whole numbers as product of their prime factors; 3. solve problems on quantitative aptitude involving LCM and HCF.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
4. Counting in Base 2	Students should be able to count in groups of twos.	Counting in groups of twos.	Guides students to prepare bundies in twos and units, and use the bundles in counting in base two.	Count with the aid of the prepared bundles.	Charts, counters such as match sticks, broom sticks bottle tops, etc	Students to in group of twos
5. Conversion of base 10 numerals to binary numbers.	Students should be able to convert base 10 numerals to binary numbers	Converting numbers 1 - 10 to base 2.	Uses bundles or piles to demonstrate conversion from base 10 to two e.g. 3 represents 1 bundles of two and 1 unit, i.e. $3_{ten} = 11_{two}$	Use bundle in twos and units to repeat teacher's demonstration for numbers 1 - 10.	Charts, counters such as match sticks, broom sticks, bottle tops.	Students to: 1. convert given numbers base 10 to numbers in base two numbers; 2. prepare conversion chart of numbers 1-10 in base 10 to base 2.

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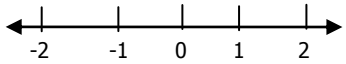
THEME: NUMBERS AND NUMERATION **JSS ONE**

SUB-THEME: FRACTIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Fractions	Students should be able to: 1. identify equivalent fractions of any given fraction; 2. Apply equivalent fractions in sharing of commodities, e.g. food, money, etc. 3. solve problems in quantitative aptitude reasoning in equivalent fractions; 4. find equivalence of any given fraction; 5. arrange given fractions either in ascending or descending order; 6. convert: ▪ fractions to decimals; ▪ decimals to fractions 7. convert: ▪ fractions to percentages; ▪ percentages to fractions	i. Identifying Equivalent fractions. ii. Quantitative aptitude reasoning. iii. Equivalent fractions iv. Ordering of fractions. v. Conversion of fractions to decimals and vice versa. vi Conversion of fractions to percentages and vice versa	1. Leads students to recognize that ½ , 2/4, 4/8, are equivalent fractions i.e. ½ = 2/4 = 4/8 2. Guides students to apply the equivalent fractions in sharing of commodities e.g. food, money etc. 3. Leads students to solve problems in quantitative aptitude involving equivalent fractions. 4. Guides students to discover the formula for obtaining an equivalent fraction of a given fraction, e.g. ½ = $\frac{1 \times n}{2 \times n}$, n = 1,2,… 5. Leads students to solve problems using the above formula. 6. Leads students to arrange given fractions in ascending or descending order of magnitude. 7. Leads students to convert: ▪ fractions to decimals; ▪ decimals to fractions. 8. Guides students to convert: ▪ fractions to percentages; ▪ percentage to fraction.	1. Recognize that ½ , 2/4, and 4/8 are equivalent fractions. 2. Apply equivalent fractions in sharing commodities e.g. food, money etc. 3. Solve problems on quantitative aptitude in equivalent fractions. 4. State and write the relationship or formula as given. 5. Solve problems using the given formula. 6. Arrange fractions in ascending or descending order. 7. Convert: ▪ fractions to decimals; ▪ decimals to fractions. 8. Convert fractions to percentages and percentages to fractions.	Charts of equivalent fractions, Charts of fractions, flash cards, Conversion Charts of percentages and fractions.	Students to: 1. solve problems on equivalent fractions; 2. solve problems on quantitative aptitude in equivalent fractions; 3. find equivalence of any given fraction, using the given formula; 4. solve problems on ordering of given fractions; 5. convert given fractions to decimals and decimals to fractions; 6. convert any fraction to percentages and vice versa.

MATHEMATICS CURRICULUM

THEME:BASIC OPERATIONS **JSS ONE**
SUB-THEME: BASIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Addition and subtraction.	Students should be able to: 1. add and subtract any given numbers correctly; 2. state the place value of each numbers in the sum or difference; 3. draw and use number line to illustrate directed numbers; 4. add and subtract positive and negative integers correctly on the number line; 5. interpret and relate positive and negative numbers to everyday activities .	i. Addition and subtraction of numbers and place values. ii. Use of number line. iii. Addition and subtraction of positive and negative integers. iv. Every day application of positive and negative integers	1. Leads students to add and subtract any two numbers up to 4-digits 2. Leads students to state the place value of each number in the sum or difference. 3. Guides students to use number line to illustrate directed numbers  4. Guides students to perform the operation of addition and subtraction of positive and negative integers on the number line. 5. Demonstrates the applications of positive and negative integers by: - Walking forward and backward; - walking up and down stairs; - reading temperatures above and below zero; - using banking deposits and withdrawals, etc.	1. Add and subtract any two numbers up to 4-digits and state the place value of the result. 2. Solve problems involving addition and subtraction of 4-digit numbers. 3. Use number line to illustrate directed numbers. 4. Perform the operation of addition and subtraction of integers on the number line. 5. Demonstrate the use of number line. 6. Solve related problems on directed numbers.	Charts / flash cards, Number line chart, Number chart, Bank statement of account, thermometer, etc.	Students to: 1. add and subtract any two given numbers of not more than 4 digits; 2. state the place value of each of the sum or difference number in the sum; 3. draw numbers line and locate numbers on the number line; 4. add and subtract positive and negative integers on the numbers line; 5. record the thermometer reading at intervals of time and different conditions; 6. walk forward and backward at specified times and relate the results to the number line.

MATHEMATICS CURRICULUM

THEME: BASIC OPERATIONS **JSS ONE**
SUB-THEME: BASIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Addition and Subtraction of fractions	Students should be able to: 1. solve given problems on addition and subtraction of fractions; 2. solve word problems involving addition and subtraction of fractions.	i. Addition and subtraction of fractions. ii. Word problems on addition and subtraction of fractions.	1. Guides the students to add and subtract fractions using diagrams and calculation. 2. Guides the students to add and subtract fractions with different denominators using diagrams and calculation. 3. Guides the students to add and subtract fractions with mixed numbers. 4. Guide the students to recognize and solve combined addition and subtraction of fraction problems. 5. Guide the students to interpret and solve word problems.	1. Add and subtract fractions using diagrams and calculation. 2. Add and subtract fractions with different denominators using diagrams and calculation. 3. Add and subtract fractions with mixed numbers. 4. Solve combined addition and subtraction of fraction problems. 5. interpret and solve word problems on combined addition and subtraction of fractions.	Fraction charts	Students to: 1. add and subtract fractions with the same denominators; 2. add and subtract fractions with different denominators; 3. add and subtract fractions with mixed numbers; 4. solve combined addition and subtraction of fraction problems; 5. solve word problems on addition and subtraction of fractions.
3. Multiplications and Divisions of fractions	Students should be able to: 1. solve problems on multiplication of fractions; 2. solve problems on division of fractions; 3. solve word problems involving multiplication and division of fractions.	i. Multiplication of fractions. ii. Division of fractions. iii. Word problems involving multiplication and division of fractions.	Guides student to: 1. Multiply and divide fractions using diagrams. 2. Multiply and divide fractions by direct calculation. 3. Multiply and divide mixed numbers by direct calculation. 4. Interpret and solve word problems involving multiplication and division of fractions.	1. Multiply and divide fractions using diagrams. 2. Multiply and divide fractions by direct calculation. 3. Multiply and divide mixed numbers by direct calculation. 4. Interpret and solve word problems involving multiplication and division of fractions.	Flash cards	Students to: 1. multiply and divide given fractions using diagrams; 2. multiply and divide the given fractions by direct calculation; 3. multiply and divide given mixed numbers by direct calculation; 4. interpret and solve given word problems involving multiplication and division of fractions.

MATHEMATICS CURRICULUM

THEME:BASIC OPERATIONS **JSS ONE**

SUB-THEME: DERIVED OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Estimation	Students should be able to: 1. estimate the dimensions and distances within the school; 2. estimate the capacity and mass of given objects; 3. estimate other things in day to day activities; 4. solve problems on quantitative reasoning in estimation.	i. Estimation of dimensions and distances. ii. Estimation of capacity and mass of objects iii. Estimation of other things eg age, time, etc. iv. Quantitative reasoning involving estimation.	1. Leads students to identify objects in the classroom and school environment and the dimensions that can be estimated. 2. Guides students to estimate some distances and dimensions. 3. Leads students to estimate the capacity of given objects. 4. Guides students to estimate the mass of given objects. 5. Guides students to estimate things in day to day activities. 6. Leads students to solve problems on quantitative reasoning in estimation.	1. Identify objects in the classroom and school environment and the dimensions that can be estimated. 2. Estimate some distances and dimensions.. 3. Estimate the capacity of some containers eg milk tins. 4. Estimate the mass of given objects. 5. Estimate the value of things in day to day activities. 6. Solve problems on quantitative reasoning in estimation.	Desks, tables, class rooms, foot paths, books, school bags, containers, milk tins, solid objects etc.	Students to: 1. estimate the lengths and widths of the class room; 2. estimate the distance between the classroom and the principal's office; 3. estimate the capacity of given objects; 4. estimate the quantitative values of other things in day –to-day life; 5. solve problems on quantitative reasoning in estimation.

MATHEMATICS CURRICULUM

THEME:BASIC OPERATIONS **JSS ONE**

SUB-THEME: DERIVED OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Approximation	Students should be able to: 1. approximate answers to addition and subtraction problems to a given degree of accuracy; 2. approximate answers to multiplication and division problems to a given degree of accuracy; 3. rounding numbers to the nearest 10, 100, and 1000; 4. apply approximation involving basic operations in everyday life activities; 5. solve problems on quantitative reasoning in the above contents.	i. Approximating values of addition and subtraction. ii. Approximating results of multiplication and division. iii. Rounding off numbers to the nearest 10; 100;and 1000. iv. Application of approximation in everyday life. v. Quantitative reasoning.	1. Leads students to approximate answers to given addition and subtraction problems. 2. Guides students to compare the actual answers with the approximated ones. 3. Guides students to approximate answers to multiplication and division problems. 4. Leads Students to round off given numbers to the nearest 10;100; and 1000. 5. Solve real life problems involving approximation. 6. Leads students to solve problems on quantitative reasoning in the above contents.	1. Carry out approximation of answers to given addition and subtraction problems. 2. Carry out the actual addition and subtraction and compare the answers with the approximated ones. 3. Carry out approximation on multiplication and division problems. 4. Carry out the actual multiplication and division and compare the actual and approximated values. 5. Round off given numbers to the nearest 10;100;and 1000. 6. Solve problems involving approximation of things in everyday activities. 7. Solve problems on quantitative reasoning in the above contents.	Addition and subtraction approximation charts. Multiplication and division approximation charts . Multiplication and division approximation charts . Rounding off number chart. Rounding off number chart.	Students to: 1. approximate answers to given addition and subtraction problems; 2. approximate answers to given multiplication and division problems; 3. round off given numbers to the nearest 10;100;and 1000; 4. solve given problems on quantitative reasoning in the above contents 5. solve problems on qualitative reasoning.

MATHEMATICS CURRICULUM

THEME:BASIC OPERATIONS **JSS ONE**
SUB-THEME: DERIVED OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Addition of numbers in base 2 numerals.	Students should be able to add two or three 3-digits binary numbers.	Addition of two or three 3-digits binary numbers.	Guides students to add two or three 3-digit numbers in base 2.	Do addition of simple two or three 3-digits binary numbers.	Counters, Sum cards.	Students to add two or three numbers involving two or three 3-digit numbers.
4. Subtraction of numbers in base 2 numerals.	Students should be able to subtract two 3-digit binary numbers.	Subtraction of two 3-digits binary numbers	Guides students to subtract two 3-digit numbers in base 2 number system.	Carry out subtraction of two 3-digit number in base 2 numbers system.	Counters	Students to subtract two numbers involving 2 –digit or 3-digits numbers.
5. Multiplication of numbers in base 2 numerals.	Students should be able to multiply two 2-digits binary numbers.	Multiplication of two 2-digits binary numbers	Guides students to multiply two 2-digt numbers in bases 2.	Carry out multiplication of 2-digit numbers in base 2.	Chart showing the multiplication of two 2-digit binary numbers.	Students to multiply given 2-digit numbers.

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS ONE**
SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Use of symbols	Students should be able to: 1. solve problems expressed in open sentences; 2. identify the relationship between addition and subtraction; multiplication and division; 3. use letters to represent symbols or shapes in open sentences.	i. Open sentences ii. Use of letters to represent symbols or shapes in open sentences	1. Guides students to find the missing number in an open statement using flash cards and open sentence charts e.g. □ - 5 = 8 7 + □ = 11 □ = 11 - 7 □ x 3 = 12 □ = 12 ÷ 3 2. Guides students to solve more problems 3. Leads Students to identify the relationship between addition and subtraction; and multiplication and division in open sentences. 4. Guides students to use letters to represent symbol e.g. 2 + □ = 11 is the same as 2 + n = 11	1. Find what the boxes represent. 2. Explain the relationship between addition and subtraction; and multiplication and division. 3. Solve more related problems. 4. Represent symbols with letters	Flash cards and open sentence charts.	Students to: 1. find what given boxes in open sentences represent; 2. explain the relationship between: (a) addition and subtraction, (b) multiplication and division, in open sentences; 3. replace the boxes in given open sentence problems with letters.

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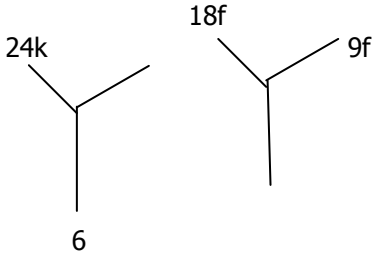
THEME: ALGEBRAIC PROCESSES **JSS ONE**
SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
Use of symbols contd.	Students should be able to: 4. solve open sentence problems involving two arithmetic operations; 5. solve word problems involving use of symbols; 6. solve quantitative aptitude problems on the use of symbols.	iii. Solving open sentences with two arithmetic operations iv. Word Problems involving use of symbols v. Quantitative aptitude.	1. Guides students to solve open sentences of the form 2 x - 1 = 7 □ □2 ÷ 2 + 3 = 8 using flash cards. 2. Guides students on how to translate word problems into mathematical expression involving symbols using Polya's principles 3. Leads students to solve quantitative aptitude problems on the use of symbols.	1. Solve open sentences problems with two arithmetic operations. 2. Solve problems on charts. 3. Translate related word problems into mathematical expressions involving symbols and solve the problems. 4. Solve quantitative aptitude problems on the use of symbols.	Flash cards and open sentence charts Flash cards and charts showing word problems expressed in symbols Flash cards and charts, etc	Students to: 4. solve open sentence problems involving two arithmetic operations; 5. translate and solve given word problems into simple expressions involving symbols; 6. solve quantitative aptitude problems on the use of symbols.

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THEME: ALGEBRAIC PROCESSES **JSS ONE**

SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Simplification of algebraic expressions	Students should be able to: 1. identify and collect like terms in a given expression; 2. identify the coefficient of a given algebraic term; 3. identify the positive and negative coefficients of a given algebraic term; 4. perform basic arithmetic operations on expressions of similar terms; 5. solve related word problems; 5. insert/remove brackets and simplify expressions; 6. solve quantitative aptitude problems on the use of brackets.	i. Like and unlike terms in algebraic expressions. ii. Identification of coefficients of terms of algebraic expressions. iii. Basic arithmetic operations applied to algebraic expressions of similar terms. iv. Collection and simplification of like and unlike terms in algebraic expressions. iv. Use of brackets. v. Quantitative reasoning.	1. Guides students to identify the coefficient of algebraic terms. 2. Leads students to identify the coefficient of positive and negative terms in algebraic expressions. 3. Guides students to perform arithmetic operations on expressions of similar terms e.g. $2x + 3x + 7x = 12x$. 4. Leads students to solve related word problems. 5. Guides students to identify and collect like and unlike terms. 6. Leads students to simplify resulting expression. 7. Guides students to remove and insert brackets from and to expressions, respectively. 8. Leads students to simplify expressions such as $(14m - 8) + (6m + 5)$. 9. Leads students to solve the problems in quantitative aptitude of the form :  	1. Identify like and unlike terms in an algebraic expression. 2. Identify coefficient of positive and negative terms in algebraic expressions. 3. Find the coefficient of positive and negative terms in algebraic expressions. 4. Perform arithmetic operations on expressions of similar terms. 5. Solve related word problems. 6. Identify and collect like terms in given algebraic expressions. 7. Simplify the resulting expression. 8. Collect like terms, insert or remove brackets and simplify the expression and Find missing terms in related quantitative aptitude problems.	Charts showing terms and their coefficients. Charts on worked examples on simplification of algebraic expressions. Charts of examples on insertion /removal of brackets. Qualitative aptitude chart.	Students to: 1. identify like and unlike terms; 2. identify the difference between coefficient of positive and negative terms in a given expression; 3. solve given word problems on simplification of algebraic expressions; 4. identify and collect like terms of given expressions and simplify resulting expression; 5. simplify given expressions involving brackets.

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS ONE**
SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Simple equations	Students should be able to: 1. translate word sentences into mathematical equations; 2. use mathematical equations to represent word sentences; 3. solve simple equations and cross-check the answers.	i. Translation of word problems into equations and vice-versa. ii. Solution of simple equations.	1. Leads students to translate word sentences into mathematical equations. 2. Leads students to use mathematical equations to represent word sentences. 3. Guides students to solve simple equations e.g. 5k + 7 = 22 and cross-check their answers.	1. Translate word sentences into mathematical statements. 2. Mention the need to use mathematical statements to represent word sentences. 3. Solve given simple equations and cross check the answers.	Word sentences charts Simple equation chart	Students to: 1. translate given word sentences into mathematical statements; 2. solve given simple equation problems.

MATHEMATICS CURRICULUM

THEME: MENSURTION AND GEOMETRY **JSS ONE**
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Plane shapes	Students should be able to: 1. state the similarities and differences between the following: square, rectangle, triangle, trapezium parallelogram and circle; 2. find the perimeter of a regular polygon: square, rectangle, triangle, trapezium, parallelogram and circle; 3. find the area of plane shapes such as squares, rectangles, parallelograms; 4. find the area of real life plane objects.	1. Similarities and differences between the following: Square, rectangle, triangle, trapezium, parallelogram and circle. 2. Perimeter of regular polygon, square, rectangle, triangle, trapezium, parallelogram and circle. 3. Area of regular plane shapes such as: Squares, rectangles, parallelograms, etc.	1. Guides students to identify the similarities and differences between the following: square, rectangle, triangle, trapezium, parallelogram and circle. 2. Guides students to determine the perimeter of each shape by (a) practical method (b) formula. 3. Guides students to find the area of the regular plane shapes (a) using graph papers (b) by formula and compare their answers . 2. Relates finding area to real life situation.	1. Identify the similarities and differences between the following: square, rectangle, triangle, trapezium, parallelogram and circle. 2. Determine the perimeter of each shape by (a) practical method (b) formula. 3. Find the area of the regular plane shapes (a) using graph papers (b) by formula and compare their answers . 4 . Relate finding area to real life situation.	Shapes of regular polygon: Square Triangle Rectangle, Parallelogram, Trapezium Circle. Graph papers	students to: 1. find the perimeter of given plane shape; 2. find the area of given regular shapes; 3. find the area of irregular shapes; 4. relate finding area to three real life situations .

MATHEMATICS CURRICULUM

THEME: MENSURTION AND GEOMETRY **JSS ONE**
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Three dimensional figures	Students should be able to: 1. identify the properties of cubes and cuboids; 2. identify the properties of pyramids and cones; 3. identity the properties of cylinders and spheres; 4. find volume of a cube and a cuboid.	i. Basic properties of cubes and cuboids. ii. Basic properties of pyramids and cones. iii. Basic properties of cylinders and spheres. iv. Volume of cubes and cuboids	1. Leads students to discover the properties of cubes and cuboids: **Cubes** • equal faces • 6 faces • 12 edges • 8 vertices **Cuboids**: • equal opposite faces • 6 faces • 12 edges • 8 vertices 2. Guides students to determine the edges, faces and vertices of pyramids and cones. 3. Guides students to discover the properties of cylinders and spheres. 4. Leads students to derive the formula for finding the volume of a cube and a cuboid. 5. Guides students to use the formula to calculate the volume of a cube and a cuboid.	1. Identify the number of faces, edges and vertices of cubes and cuboids' and that cubes have equal faces while opposite faces of a cuboids are equal. 2. Determine the number of edges, faces and vertices of pyramids and cones. 3. Identify the properties of cylinders and spheres. 4. Derive the formula for finding the volume of a cube and a cuboid. 5. Use the formula to calculate the volume of a cube and a cuboid.	Cubes, Cuboids, Ruler, Tapes, Empty carton, Bricks etc. Pyramid Cone. Cylinder and sphere (standard or improvised) Cubes Cuboid.	Students to: 1. identify the number of faces and vertices of a given cube; 2. identity the number of edges and faces of a cuboids; 3. identify the number of faces, edges and vertices of a given pyramid and cone; 4. identify the properties of a given cylinder and sphere; 5. find volume of a given cube and cuboid.

MATHEMATICS CURRICULUM

THEME: MENSURTION AND GEOMETRY **JSS ONE**
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Construction	Students should be able to: 1. construct parallel and perpendicular lines; 2. bisect a given line segment; 3. construct angles 90 and 60 degrees.	i. Construction of parallel and perpendicular lines. 2. Bisection of a given line segment. 3. Construction of angles 90 and 60 degrees.	Guides Students to: i. construct parallel and perpendicular lines; ii. bisect a given line segment; iii. construct angles 90 and 60 degrees.	1. Construct parallel and perpendicular lines. 2. Bisect a given line segment. 3. Construct angles 90 and 60 degrees.	Plane sheets of paper and mathematical set.	Students to: 1. construct given parallel lines; 2. construct given perpendicular lines
4. Angles	Students should be able to: 1. measure angles; 2. identify vertically opposite, adjacent, alternate and corresponding angles; 3. state properties of angles; 4. identify angles at a point and angles on a straight line and state their properties.	i. Measurement of angles. ii. Identification and properties of: ▪ vertically opposite, ▪ adjacent, ▪ Alternate, and ▪ Corresponding angles. iii. identification and properties of angles at a point and angles on a straight line	1. Leads students to measure angles. 2. Guides students to: i. identify and state the properties of the different types of angles; ii. relates the angles to real life situations. 3. Leads students to identify angles at a point and angles on straight line and state their properties.	1. Measure some given angles. 2. identify and state the properties of the different types of angles from the appropriate diagrams. 3. Relate angles to life situation. 4. Identify angles at a point and angles on a straight line and state their properties.	Protractor, plane sheet, cardboard containing angles Pencil, Metre rule Angles, Charts of angles at a point and angles on a straight line.	Students to: 1. measure given angles; 2. identify vertically opposite, adjacent, alternate and corresponding angles from given diagrams; 3. state the properties of given angles; 4. relate angles to three life situations; 3. identify angles at a point and angles on a straight line in given diagrams.

MATHEMATICS CURRICULUM

THEME: EVERYDAY STATISTICS **JSS ONE**

SUB-THEME: DATA COLLECTION AND PRESENTATION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Need for statistics	Students should be able to: 1. list purposes of statistics; 2. recognize the usefulness of statistics for planning purposes; 3. apply the occurrence of chance events / application of probabilities in everyday life; 4. recognize the usefulness of statistics for prediction purposes.	i. Purposes of statistics ii. Need for collecting data for planning purposes Collection of data	1. Leads students in discussing the purposes of statistics. 2. Introduces Students to the meaning of population, drug abuse, voter education and environmental education. 3. Leads students in discussing the usefulness of statistics for planning purposes. 4. Leads students to discuss usefulness of the data collected from voter education, consumer education. 5. Guides students to apply the probability/ occurrence of chance events in everyday life. 6. Leads students to discuss the usefulness of statistics such as from drug abuse, environmental education, HIV/AIDS etc. for prediction purposes.	1. Discuss the purposes of statistics. 2. State the meaning of drug abuse, voter education and environmental education etc. 3. Discuss the usefulness of statistics for planning purposes. 3. Source for information from relevant agencies 4. Mention the application of probability of chance and events in everyday life. 5. Discuss the usefulness of statistics for prediction purpose.	1. Chart of purposes of statistics 2. Charts of information on drug abuse, voter education environmental education etc. 3. usefulness of statistics for planning purposes. 4. Probability charts 5. Charts of usefulness of statistics for prediction purposes	Students to: 1. list three purposes of statistics; 2. list some situations that need data collection for planning purposes; 3. give four examples of the application of probability in everyday life; 4. list situations around him/her where statistics is used for prediction.
2. Data collection	Students should be able to collect data in the class	Collect data in the class	Guides students to collect data in the class	Collect data in the class	Record of Students bio data, marks, score sheets, dice coins.	Students to collect Bio-data on Students with given characteristics
3. Data presentation	Students should be able to determine the median of a given set of data	Median	1. Guides students to define median 2. Guides students to find the median of a given set of data such as ▪ from odd numbered data ▪ from even numbered data.	1. Define median 2. Find the median of a given set of data.	Median charts.	Students to find the median of a given set of data.

MATHEMATICS CURRICULUM

THEME: NUMBERS AND NUMERATION **JSS TWO**

SUB-THEME: WHOLE NUMBERS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Whole numbers	Students should be able to: 1. express any whole number in standard form; 2. express decimal number in standard form; 3. find the prime factors of numbers not greater than 200; 4. express numbers as products of its prime factors; 5. find the least common multiples of numbers (LCM) 6. find the highest common factor (HCF) of numbers; 7.a) identify numbers that are perfect Squares; 7.b) find squares of any given numbers; 7.c) find the Square root of perfect squares using factor method; 7.d) find square root of any given number; 8. solve quantitative reasoning problems related to contents i – vii.	i. Whole numbers in standard form. ii. Decimal numbers in standard form. iii. Prime factors. iv. Prime factorization. v. LCM vi. HCF vii. Squares and square roots. viii. Quantitative reasoning.	1. Leads students to express any given whole number in standard form. i.e. $a \times 10^n$, $1 \leq a < 10$, $n \geq 0$. 2. Leads students to express decimal numbers in standard form. i.e. $a \times 10^n$, $1 \leq a < 10$, $n \leq 0$ 3. Guides students to find prime factors of numbers 4. Guide students to express numbers as products of its prime factors 5. Guides students to find the LCM of numbers 6. Leads students to find the highest common factors of numbers. 7. Guide students to identify numbers that are perfect square. 8. Guide students to find the squares of any given no. 9. Guides students to find square roots of perfect square by factor method. 10. Lead students to find square root of any given number. 11. Guides students on how to solve quantitative reasoning problems on contents i – vii .	1. Express any whole number in standard form. 2. Express decimal numbers in standard form. 3. Find prime factors of numbers. 4. Express numbers as products of its prime factors. 5. Find the LCM of numbers. 6. Find the highest common factor of any given whole number. 7. Identify numbers that are perfect square. 8. Find the squares of any given numbers. 9. Find the square root of perfect squares by factor method. 10. Find the square root of any given number. 11. Solve problems on quantitative reasoning related to contents i – vii	Flash cards Flash cards Flash Card	Students to: 1. express given whole numbers in standard form; 2. express given decimal numbers in standard form; 3. find prime factors of given numbers not greater than 200; 4. express given numbers as products of its prime factors; 5. find the least common multiples (LCM) of given numbers; 6. find the highest common factors of given numbers 7a. find the square of any given number; 7b. find the square roots of given perfect squares using the factor method; 7c. find the square root of any given number; 8. solve given problems on quantitative reasoning related to content i – vii.

THEME: NUMBERS AND NUMERATION **JSS TWO**
SUB-THEME: FRACTION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Fractions	Students should be able to: 1. convert simple fractions to ratios, decimals and percentages and vice versa; 2. solve quantitative reasoning problems related to conversion of fractions to ratios, decimals and percentages;	i. Expressing fractions as ratios, decimals and percentages. ii. Quantitative reasoning on fractions, ratios and percentages.	1. Guide students to convert fractions to ratios, decimals and percentages and vice and versa. 2. Guides students to solve quantitative reasoning problems related to conversion of fractions to ratios, decimals and percentages and vice versa.	1. Convert fractions to ratios, decimals and percentages and vice and versa solve quantitative reasoning problems related to conversion of fractions to ratios, decimals and percentages	Squared paper or graph sheet; Flash Cards	Students to: 1. convert given fractions to ratios, decimals and percentages; 2. solve given problems on Quantitative reasoning related to the contents.

MATHEMATICS CURRICULUM

THEME: BASIC OPERATIONS **JSS TWO**

SUB-THEME: DERIVED OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	Students		
1. Transactions in the homes and offices	Students should be able to: 1. solve problems relating to office and household arithmetic; 2. solve simple commercial arithmetic relating to profit, interest, discount and commission.	i. Household arithmetic. ii. Commercial arithmetic.	1. Leads students to solve common home and office problems e.g. electricity bills, water rates and family budget. 2. Guides students to solve simple commercial arithmetic relating to profit, interest, discount and commission.	1. Solve problems on electricity bills, water rates and family budget. 2. Solve given simple commercial arithmetic.	Electricity bill water rate bill	Students to: 1. solve problems on household and office arithmetic; 2. solve given problems on commercial arithmetic.
2. Approximation	Students should be able to: 1. approximate numbers to any given degree of accuracy; 2. solve quantitative reasoning problems related to approximation of numbers.	1. Approximation of numbers: • Decimal places • Significant figures ii. Quantitative reasoning	1. Leads students to carryout simple approximation to a given degree of accuracy. 2. Guides students to solve quantitative reasoning problems related to approximation of numbers.	1. Carryout simple approximation to a given degree of accuracy. 2. Solve quantitative reasoning problems related to approximation of numbers.		Students to: 1. solve simple problems relating to: i. number of given decimal places; ii. the required significant figures. 2. solve quantitative reasoning problems related to approximation of numbers.
3. Multiplication and division of directed numbers	Students should be able to: 1. obtain the squares and square roots of numbers; 2. interpret and use tables, charts, records and schedules; 3. carryout correct multiplication and division involving directed number.	i. Square and square root tables. ii. Charts, records and schedules. iii. Multiplication and Division of directed numbers.	1. Leads students to obtain the squares and square roots of numbers from the table. 2. Guides students to interpret and use tables, charts, records and schedules. 3. Leads students to multiply and divide directed numbers.	1. Obtain the squares and square roots of given numbers form the table. 2. Interpret and use tables, charts, records and schedules. 3. Multiply and divide directed numbers.	Square and square root tables. Distance charts, fight schedules charts etc. Directed numbers multiplication and division chart.	Students to: 1. obtain the squares and square roots of given numbers from the table; 2. interpret and use given tables, charts, records and schedules; 3. find the product of any two given directed numbers; 4. determine the quotient of any two given directed numbers.

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS TWO**

SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Algebraic expressions	Students should be able to: 1. expand a given algebraic expression; 2. factorize simple algebraic expressions; 3. apply the use of quadratic equation box in expanding and factorizing algebraic expressions; 4. solve quantitative reasoning problem; 5. simplify algebraic expression on fractions with monomial denominators; 6. interpret and solve word problems involving algebraic fractions.	i. Expansion of algebraic expression. ii. Factorization of simple Algebraic expressions. iii. Expansion and factorization of quadratic expressions. iv. Quantitative reasoning v. Algebraic expressions of fractions with monomial denominators. vi. Word problems leading to simple algebraic fractions	1. Leads Students to expand algebraic expression of the form **a(b+c) and (a + b)x (c+d**) 2. Guides students to factorize algebraic expressions. 3. Leads students to use a quadratic equation box in expanding and factorizing quadratic expressions. 4. Guides students to find missing factors in samples of the form $x^2 + 4x + 3$ $x + 1$ | ☐ 5. Guides students to simplify algebraic expansion of the form $\frac{1}{p} + \frac{1}{q} = \frac{P+q}{pq}$ 6. Leads students to interpret word problems leading to algebraic fractions. 7. Guide students to solve the resulting algebraic fractions.	1. Expand given algebraic expressions. 2. Factorize given algebraic expressions. 3. Use quadratic equation box in expanding and factorizing quadratic expressions. 4. Find the missing factor in the sample. 5. Simplify algebraic expression of fractions with monomial denominators. 6. Interpret word problems leading to algebraic fractions. 7. Solve the resulting algebraic fractions.	Flash cards Flash cards Quadratic equation box Flash cards Flow charts	Students to: 1. expand given algebraic expressions; 2. factorize given algebraic expressions; 3. expand and factorise quadratic expressions; 4. solve quantitative aptitude problem; 5. simplify given algebraic expressions with monomial denominator; 6. interpret and solve given word problems leading to algebraic fraction;

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS TWO**

SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Simple equations	Students should be able to solve problems on simple equations	Problems on simple equations	Guides students to solve problems on simple equations such as: $3n - 4 = 2n + 1$ $\frac{2y + 4}{7} = 2$	Solve problems on simple equations	Flash cards	Students to solve given problems on simple equations
3. Linear inequalities	Students should be able to: 1. identify linear inequality in one variable; 2. solve linear inequality in one variable; 3. represent solutions of linear inequalities in one variable on number line; 4. Solve word problems involving linear inequalities in one variable.	i. Linear inequality in one variable. ii. Graphical representations of solutions of linear inequalities in one variable. ii. Word problems	1. Guides students to identify linear inequalities in on variable. 2. Leads students to solve linear inequality in one variable. 3. Guides students to solve and represent the solutions of linear inequalities on the number line. 4. Leads students to interpret and solve word problems involving linear inequalities in one variable.	1. Identify linear inequalities in one variable. 2. Solve linear inequalities in one variable. 3. Solve and represent inequalities on number lines. 4. Interpret and solve word problems involving linear inequalities in one variable.	Flash cards Cardboard paper , graph board	Students to: 1. identify given linear inequality 2. solve given problems on linear inequality in one variable; 3. solve given problem on linear inequality; 4. represent the solution on the number line; 5. interpret & solve word problems involving linear inequalities in one variable.

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS TWO**

SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Graphs	Students should be able to: 1. identify x–axis and y –axis; 2. plot points on the Cartesian plane; 3. Prepare table of values; 4. Plot the graph of linear equations in two variables; 5. interpret the plotted graph; 6. plot linear graphs from real life situations; 7. solve quantitative aptitude problem.	i. Plotting points on the Cartesian plane. ii. Graph of Linear equation in two variables. iii. Linear graphs from real life situations. iv. Quantitative reasoning.	1. Leads students to identify ¯x and ¯y axes. 2. Guides students to plot given points on Cartesian plane. 3. Leads students to compile table of values of linear equation in two variables. 4. Guides students to plot given points on Cartesian plane. 5. Leads students to interpret information presented on the graph. 6. Plot the graph of the table of values. 7. Guides students to plot linear graphs from real life situations e.g. Distance – Time graph Velocity – Time graph Item – price graph	1. Identify x and y axis on the Cartesian plane. 2. Plot given points on Cartesian plane. 3. Compile tables of values of given linear equations in two variables. 4. Plot the graph of given linear equation in two variables. 5. Interpret information presented on the graph. 6. Plot a linear graph from given real life situation.	Graph board, graph paper, rule, Pencil. Distance time data, Velocity – Time data, etc. Table of values	Students to: 1. identify x and y axis on the Cartesian plane; 2. plot given points on Cartesians plane; 3. compile table of values of given linear equation in two variables; 4. plot the graph of given linear equation in two variable; 5. plot linear graph from given real life situation.

MATHEMATICS CURRICULUM

THEME: MENSURATION AND GEOMETRY **JSS TWO**
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Plane figures/ Shapes	Students should be able to: 1. state the properties of parallelogram, rhombus and kite; 2. identify these shapes in their environment; 3. draw plane objects to scale; 4. convert actual length to scales and vice versa; 5. apply scale drawing to solve measurement problem; 6. solve problems on quantitative aptitude related to plane shapes/figure and scale drawing	i. properties of parallelogram, rhombus and kite. ii. Scale drawing.	1. Guides students to identify the properties of parallelogram, rhombus and kite. 2. Lead students to identify these shapes in the environment, 3. Guides students to measure actual lengths of teachers table and their classroom. 4. Leads students to convert actual length to given scale and vice versa. 5. Leads students to apply scale drawings in solving measurement problems. 6. Guides students to solve quantitative aptitude problems related to plane shape/figure and scale drawings.	1. Identity the properties of parallelogram, rhombus and kite. 2. Identify these shapes in the environment. 3. Measure the actual length of the given objects and draw them to scale. 2. Convert actual length to scales and vice versa. 3. Apply scale drawing to solving measurement problems. 4. Solve quantitative aptitude problems on plane shapes and scale drawings.	Models of parallelogram, rhombus and kite Tape (steel and rolling) Ruler Pencil Plain sheets.	Students to: 1. state the properties of the following: • Rhombus • Parallelogram • kite 2. draw actual length to scale; 3. solve problems on quantitative aptitude related to plane shapes/figures and scale drawing.

MATHEMATICS CURRICULUM

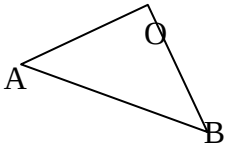
THEME: MENSURATION AND GEOMETRY **JSS TWO**

SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. Angles	Students should be able to: 1. find the sum of angles of triangles; 2. find sum of angles in a quadrilateral; 3. find the sum of the interior angles of a polygon; 4. distinguish between angles of elevation and angles of depression; 5. use angles of elevation and depression in calculating distances and heights using scale drawing; 6. identify the relationship between angles of elevation and depression; 5. solve problems on quantitative aptitude related to angles.	i. Sum of angles of a polygon. ii. Angles of elevation and depression	1. Guides students to find the sum of angles of triangles. 2. Guides student to find sum of angles in a quadrilateral. 3. Leads students to discover that the sum of interior angles of an **n-sided** convex polygon is **2n– 4 right angles.** 4. Leads students to identify what angles of elevation and depression means. 5. Guides students to use angles of elevation and depression in calculations using scale drawing. 6. Guides students to identify the relationship between angles of elevation and depression. 7. Leads students to solve problems on quantitative aptitude related to angles.	1. Find the sum of angles of triangles. 2. Find sum of angles in a quadrilateral. 3. Use the sum of interior angles to calculate any given related problem. 4. Identify what angles of elevation and depression mean. 5. Use angles of elevation and depression in calculations using scale drawing. 6. Identify the relationship between angles of elevation and depression. 7. Leads students to solve problems on quantitative aptitude related to angles.	Cut out shapes of triangles, Quadrilateral, and other polygon types. Rulers, tapes Protractor.	Students to: 1. find the sum of angles of triangles; 2. find sum of angles in a quadrilateral; 3. find the sum of interior angles of a given convex polygon; 4. use angles of elevation and depression to calculate distances and heights; 5. solve problems on quantitative aptitude related to angles.

MATHEMATICS CURRICULUM

THEME: MENSURATION AND GEOMETRY **JSS TWO**
SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Bearing	Students should be able to: 1. identify the cardinal points; 2. locate the position of objects; 3. find distances between objects using scale drawing.	1. Bearing	1. Guide students to identify the major and minor cardinal points. 2. Guide students to locate position of objects using: -3 digit bearing -Compass bearing. 3. Guide students to find distances between objects using scale drawing.	1. Identify the major and minor cardinal points. 2. Locate the position of objects using -3 digit bearing -Compass bearing 3. Find distances between objects using scale drawings.	Protractor Ruler, compass 360^0 – (wooden, metal, plastic or cardboard), clinometer.	Students to: 1. mention: a. the major cardinal points; b. The minor cardinal points. 2. what is the bearing of A from B, B from A  3. using scale drawing, find the distance of B from A.
4. Construction	Students should be able to: 1. Construct triangles given: (a)2 sides and a secluded angle (b)2 angles and a side between them (c) All the 3 sides 2. Bisect any given angles	1.Constructing triangles. 2.Bisecting angles.	1. Guide students to construct triangles using a protractor and ruler 2. Lead student to bisect any given angle using a pair of compasses, ensuring that students follow the basic steps	1. Construct triangles using protractor and ruler. 2. Bisect any given angle following the given basic steps.	1. Plain paper 2. Protractor 3. Ruler 4. Pair of compasses, mathematical set	Students to: 1. construct triangles giving: (a) 2 sides and a secluded angle; (b) 2 angles and a side between them; (c) All the 3 Sides. 2. bisect any given angles.

MATHEMATICS CURRICULUM

THEME: EVERYDAY STATISTICS **JSS TWO**

SUB-THEME: DATA COLLECTION AND PRESENTATION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Data presentation	The Students should be able to: 1. present data in an ordered form; 2. construct frequency tables from any given data; 3. draw pie charts; 4. read information from pie chart; 5. generate and use data for statistical purposes; 6. interpret and use tables, charts, records and schedules.	i. Ordered presentation of data ii. Frequency table. iiii. Pie charts iv. Charts, records and schedules	1. Guide students to arrange data in an ordered form. 2. Guides students to display any given data on frequency table. 3. Guides students to draw pie charts. 4. Guides students to read information from pie chart. 5. Leads students to generate and use data for statistical purposes. 6. Guides pupil to interpret and use tables, charts, records and schedules.	1. Arrange data in an ordered form. 2. Display any given data on frequency table. 3. Draw pie charts. 4. Read information from pie chart. 5. Generate and use data for statistical purposes. 6. Interpret and use tables, charts, records and schedules.	Source information : Distance charts, flight schedules etc.	Students to: 1. present data in ordered form; 2. construct frequency table from a given data; 3. draw a pie chart; 4. read information from given pie charts.
2. Probability	Students should be able to: 1. discuss the occurrence of chance events in everyday life; 2. determine the probability of certain events; 3. apply the occurrence of chance events / probabilities in everyday life.	i. Occurrence of chance events in everyday life. ii. Probability of chance events.	1. Leads students to give examples of chance events in everyday life e.g. The chance or event that rain will fall in December. 3. Leads students to perform experiments using coin and die. 4. Guides students to calculate the probability of the resulting experiments. 5. Guides students to analyze statistical data with the use of probability such as mortality rate etc. 6. Guides students to apply the probability/ occurrence of chance events in everyday life.	1. List examples of chance events in everyday life 2. Perform experiments using coin and die. 3. Calculates the probability of the resulting experiments 4. Analyze statistical data with the use of probability 5. Mention the application of probability in everyday life.	Coins Dice Source-information-sheet on events	Students to: 1. Give three occurrence of chance events 2. Calculate the probability of chance events. 3. Give four examples of the application of probability in everyday life.

MATHEMATICS CURRICULUM

THEME: NUMBERS AND NUMERATION **JSS THREE**
SUB-THEME: WHOLE NUMBERS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Whole numbers	Students should be able to: 1. revise basic operations in binary system; 2. convert numbers in binary numbers to other bases and vice versa; 3. solve quantitative aptitude problems on Binary number systems; 4. use computer to do simple mathematical calculations ; 5. translate word problems into numerical expression.	i. Binary number system iii. Using computer to do simple mathematical calculations iv. Translation of word problems into numerical expressions	1. Guides students to revise the basic operations in binary number system using electrical switch 2. Guides students to convert numbers in binary system to other bases and vice versa. 3. Guides students to solve quantitative aptitude involving binary numbers system 4. Guides the students to do some simple calculations on computer 5. Guides students to interpret each word problem into numerical expression e.g. the difference between 7 and the sum of half of 8 and 3	1. Carryout operation in binary number system 2. Convert numbers in binary system to other bases and vice versa. 3. Solve quantitative Aptitude involving binary number system 4. Do some simple calculations on computer 5. Interpret each word problem into numerical expression	Computer system, flash cards.	Students to: 1. solve problems involving basic operations in binary number system; 2. convert given numbers in binary number system to other bases and vice-versa; 3. solve quantitative Aptitude problems; 4. use computer to solve given mathematical problems; 5. interpret and solve each given word problem into numerical expression;

THEME: NUMBERS AND NUMERATION
SUB-THEME: WHOLE NUMBERS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
Whole numbers contd.	Students should be able to: 6. simplify expressions involving brackets and fractions; 7. solve problems involving direct and inverse proportions; 8. apply direct and inverse proportions to practical problems; 9. solve problems on compound interest; 10. apply the use of compound interest in daily life activities.	v. Expressions involving brackets and fractions. vi. Direct and inverse proportion. vii. Application of direct and inverse proportions. viii. Compound interest.	1. Guides students to simplify expressions involving brackets and fractions. 2. Leads students to solve problem on direct and inverse proportion. 3. Guides students to solve practical problems in direct and inverse proportions e.g. problems on speed, productivity, consumption and reciprocals. 4.Guides students to revise simple interest 5. Leads students to solve problems on compound interest. 6. Leads students to the use of compound interest in daily life e.g. in mortgages and loans and pension	1. Simplify expression involving brackets and fractions. 2. Solves problems on direct and inverse proportion 3. Solve practical problems on direct and inverse proportion. 4. Revise simple interest. 5. Solve problems on compound interest. 6. Apply the use of compound interest in daily life.	Flash cards Direct and inverse proportion chart Source for relevant information on inverse/direct proportion Source for information on compound interest e.g. fixed deposits	Students to: 6. simplify given expressions involving brackets and fraction; 7. solve given problem on direct and inverse proportion; 8. solve given practical problem using direct and inverse proportion; 9. solve given problems on compound interest; 10. list four real life situations where compound interest apply.
2. Rational and non-rational numbers	Students should be able to: 1. identify rational and non-rational numbers.	i. rational and non - rational numbers.	1. Guides students to identify rational and non rational numbers. 2. Guides students to determine practically the approximate value number pi.	1. Identify rational and non rational numbers. 2. Determine practically the approximate value of number pi.	Thread or string Table rule Cylindrical shaped objects	Pupils to: 1. identify rational and non-rational numbers; 2. determine practically the approximate value of number pi.

MATHEMATICS CURRICULUM

THEME: BASIC OPERATIONS **JSS THREE**
SUB-THEME: BASIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Addition of numbers in base 2 numerals.	Students should be able to add two or three 3-digit binary numbers.	Addition of two or three 3-digit binary numbers.	Guides students to add two or three 3-digit numbers in base 2.	Do addition of simple two or three 3-digit binary numbers.	Counters, Sum cards.	Add two or three numbers involving two or three 3-digit numbers.
2. Subtraction of numbers in base 2 numerals.	Students should be able to subtract two 3-digit binary numbers.	Subtraction of two 3-digit binary numbers	Guides students to subtract two 3-digit numbers in base 2 number system.	Carry out subtraction of two 3-digit numbers in base 2 numbers system.	Counters	Subtract two numbers involving 2-digit or 3- digit numbers.
3. Multiplication of numbers in bases 2 numerals.	Students should be able to multiply two 2-digit binary numbers.	Multiplication of two 2-digit binary numbers	Guides students to multiply two 2-digt numbers in bases 2.	Carry out multiplication of 2-digit numbers in base 2.	Chart showing the multiplication of two 2-digit binary numbers.	Multiply given 2-digit numbers.
4. Division of numbers in base 2 numerals	Students should be able to divide two 2-digit binary numbers.	Division of two to 3-digit binary numbers.	Guides students to divide two to 3-digit binary numbers.	Carry out division of two to 3-digit binary numbers.	Charts showing the division of two 2-digit binary numbers	Divide given two 3-digit binary numbers.

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS THREE**

SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Factorization	Students should be able to: 1. factorize simple algebraic expressions 2. factorize quadratic algebraic expressions using quadratic equation box 3. solve word problems involving factorization	i. Factorization of expressions of the form: ax + ay 3m+ pq+3p +mp $a^2 - b^2$ $a^2 - 2ab + b^2$ ii. Word problems involving factorization.	1.Guides students to factorize the simple algebraic expressions 2. Leads students to use the quadratic equation box to factorize quadratic expressions 3. Guides students to translate word problem into algebraic expression 4. Leads students to factorize resulting algebraic expression	1. Factorize the given simple algebraic expressions. 3. Use the quadratic equation box to factorize quadratic expressions. 1. Translate given word problems into algebraic expression. 2. Factorize the resulting algebraic expression.	Quadratic equation box Flash cards of word problems	Students to: 1. factorize given algebraic expressions; 2. factorize given quadratic expressions using quadratic equation box; 3. translate given word problems into algebraic expressions and factorize.
2. Simple equations involving fractions	Students should be able to: 1. solve simple equations involving fractions; 2. solve simple word problems on simple equations involving fractions.	iii. Simple equations involving fractions. iv. Word problems leading to simple equation involving fractions.	1. Leads students to solve problems on simple equations involving fractions. 2. Guides students to translate word problem leading to simple equations with fractions. 3. Leads students to solve the resulting algebraic equation	1. Solve given problems on simple equations involving fractions. 2. Translate each word problem to algebraic equation 3. Solve the resulting equation	Flash cards of simple equations showing fractions Flash cards of simple squares involving fractions.	Students to: 1. solve given problems on simple equations involving fractions; 2. translate given word problem into algebraic equation and solve them.

MATHEMATICS CURRICULUM

THEME: ALGEBRAIC PROCESSES **JSS THREE**

SUB-THEME: ALGEBRAIC OPERATIONS

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Simultaneous linear equations	Students should be able to: 1. compile table of values for simultaneous linear functions; 2. solve problems involving linear simultaneous equations in two variables graphically; 3. solve simultaneous linear equations in two variables using elimination method; 4. apply elimination method to solve problems involving real life activities; 5. solve simultaneous linear equations in two variables using substitution method; 6. apply substitution method to solve problems involving real life activities.	i. Compilations of table of values. ii. Graphical solution of simultaneous linear equations in two variables. i. Solution of simultaneous linear equations using elimination method. iv. Solution of simultaneous linear equations using substitution method	1. Guides students in calculating the values of a linear equations for different values of the variables and present their solutions in a table. 2. Guides students to plot the points for the two linear equations on the same axes 3. Leads students to locate and read points of intersection of the two graphs as the solution of the equations. 4. Leads students to solve simultaneous linear equations in two variable using elimination method 5. Guide students to apply elimination method to solve problems involving real life activities. 6. Leads students to solve simultaneous linear equations in two variables using substitutions methods 7. Guides students to apply substitution method to solve problems involving real life activities.	1. Compute linear equations for different values for the variable. 2. Present calculated values in tabular form. 3. Plot the points for the two linear equations on the same axes. 4. Locate and read points of intersection of the two graphs as the solution of the equations. 5. Solve given simultaneous linear equations using elimination method. 6. Apply elimination method to solve problems involving real life activities. 7. Solve given simultaneous linear equations using substitution method. 8. Apply substitution method to solve problems involving real life activities.	Table of values Ruler, Graph board, Pencil, Flash cards with simultaneous linear equations.	Students to: 1. compute table of values for given linear functions; 2. solve a given simultaneous equations in two variables graphically; 3. solve given simultaneous linear equation using elimination method; 4. solve given simultaneous linear equations using substitution method.

MATHEMATICS CURRICULUM

THEME: MENSURATION AND GEOMETRY **JSS THREE**

SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Similar Shapes	Students should be able to; 1. identify similar figures:- triangles rectangles, squares cubes and cuboids; 2. identify the presence of similar shapes in the environment; 3. enlarge figures using scale factors; 4. calculate lengths, areas and volumes of similar figures; 5. Solve problems on quantitative reasoning involving similar shapes.	i. Similar shapes. ii. Enlargements and scale factor. iii. Lengths, areas and volumes of similar figures.	1. Guides students to identify similar shapes around them 2. Introduces the meaning of enlargement and scale factor. 3. Guides Students to enlarge shapes using scale factor. 4. Leads students to identify that if the length of similar figures are in ratio **1:k** then areas and volumes are in ratios **$1:k^2$** and **$1:k^3$** respectively. 5. Leads students to solve quantitative aptitude problems involving similar shapes.	1. Identify similar shapes around them 2. Describe the meaning of enlargement and scale factor 3. Enlarge shapes using scale factors 4. Identify that if the length of similar figures are in ratio **1:k**. then the areas and volumes are in ratios **$1:k^2$** and **$1:k^3$** respectively 5. Solve problems on quantitative aptitude involving similar shape.	Similar shapes of triangles, rectangles, squares, cubes, cuboids	Students to: 1. identify Given similar shapes; 2. determine the scale factors of any given figures; 3. enlarge figures using the scale factors**;** 4. find the lengths, areas and volume of given similar figures; 5. solve problems on quantitative aptitude involving similar shape.
2. Trigonometry	Students should be able to: 1. identify sine, cosine and tangent of an acute angle; 2. solve problems on applications of trigonometric ratios to finding distances and lengths; 3. apply trigonometric ratios in solving world problems.	i. The sine, cosine and tangent of an acute angle. ii. Applications of Trigonometric ratios.	1. Leads students to observe the sine, cosine, and tangent of an acute angle as ratios of sides of right angled triangle. 2. Leads students to use the trigonometric ratios to solve practical problems involving right-angled triangle. 3. Leads students to apply trigonometric ratios to solve world problems.	1. Identify sine, cosine and tangent of an acute angle as ratios of sides of a right-angled triangle. 2. Use trigonometric ratio to solve practical problems. 3. Apply trigonometric ratios to solve world problems.	Model of right angled triangle Flash cards with different problems.	Students to: 1. determine sine, cosine and tangent of an acute angle from given triangles; 2. solve practical problems involving trigonometric ratios.

MATHEMATICS CURRICULUM

THEME: MENSURATION AND GEOMETRY **JSS THREE**

SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. Area of plane figures	Students should be able to: 1. find areas of triangles; 2. find area of a parallelograms; 3. find areas of trapezium; 4. find area of circles and sectors; 5. Solve word problems involving areas; 6. solve quantitative aptitude problems on areas.	i. Area of triangles. ii. Area of parallelogram. iii. Area of trapezium. iv. Areas of circles. v. Word problems involving area.	1. Leads students to derive formula for areas of triangles. 2. Guides students to use the obtained formula to find the area of triangle. 3. Leads students to derive the formula for area of parallelogram. 4. Guides students to use the derived formula to find the area of parallelogram. 5. Leads students to derive the formula for area of trapezium. 6. Guides students to solve problems on area of trapezium using the derived formula. 7. Leads students to derive the formula for area of circles and the sectors. 8. Guides students to use the derived formula to find the area of circles and sectors. 9. Guides students to interpret and Solve world problems involving areas. 10. Guide students to solve quantitative aptitude problem an area of figures.	1. Derive formula for areas of triangles. 2. Use the formula to find the area of triangle. 3. Derive the formula for area of parallelogram 4. Use the derived formula to find the area of a parallelogram. 5. Derive the formula to find the area of a trapezium. 6. Find the area of trapezium. 7. Derive the formula for area of circles and sectors 8. Use the derived formula to find the area of circles. 9. Interpret and solve world problems involving areas . 10. Solve quantitative aptitude problems on areas.	Triangular shapes Models of parallelogram Models of trapezium Model of circles and sectors Flash cards with word problems.	Students to: 1. find the area of given triangles; 2. find the area of given parallelograms. 3. find areas of given trapeziums; 4. find the areas of given circles and sectors; 5. solve given world problems involving area .

MATHEMATICS CURRICULUM

THEME: MENSURATION AND GEOMETRY **JSS THREE**

SUB-THEME: SHAPES

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
5. Construction	Student should be able to: 1. construct angle 45° 2. construct angles 30^{0} 3. use a pair of compasses to copy a given angle 4. construct simple shapes.	iv. Construction of angle 45^{0} v. Construction of angles 30^{0} vi. Copying given angles v. Construction of simple plane shapes.	1. Leads students to construct angles 45^{0} 2. Leads students to construct angles 30^{0} 3. Leads students to copy a given angle 4. Guides students to Construct simple plane shapes.	1. Construct angles 45^{0} 2. Construct angles 30^{0} 3. Copy a given angle 4. Construct simple plane shapes	Mathematical set.	Students to: 1. construct angle 45^{0} 2. construct angles 30^{0} 3. copy a given angle; 4. construct simple plane shapes.

MATHEMATICS CURRICULUM

THEME: EVERYDAY STATISTICS **JSS THREE**

SUB-THEME:DATA COLLECTION AND PRESENTATION

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING RESOURCES	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. Measure of central tendency	Students should be able to: 1. review their previous work on mean, mode and median; 2. calculate the median of a given data; 3. find the mode of given data; 4. calculate the mean of any given data; 5. find the range of any given data; 6. apply measures of central tendency to analyze any given information.	i. Revision of previous work on mean, median and mode ii. Median iii. Mode iv. Mean v. Range vi. application of measures of central tendency to analyze any given information	1. Guides students to review the previous work on mean, median and mode of data. 2. Guides students to find the median of ungrouped data. 3. Guides students to find the mode of ungrouped data. 4. Leads students to calculate the means of a given data 5. Leads students to find the range of any given data. 6. Leads students to analyze any given data using the measures of central tendency.	1. Review the meaning of mean, median and mode of data. 2. Find the median of ungrouped data. 3. Find the mode of ungrouped data. 4. Calculate the mean of a given data. 5. Find the range of any given data. 6. Analyze given data using the measures of central tendency.	Demographic data Source data from Federal /State/Local Government Agencies	Students to: 1. carryout review exercises on mean, mode and median; 2. find the median of a given set of data 3. find the mode in any given data; 4. calculate the mean of any given data; 5. find the range of any given data; 6. use measures of central tendency to analyze any given data.
2. Data presentation	Students should be able to represent and interpret an information on pie charts	Pie Charts	Leads students to represent and interpret data in pie charts	1. Represent and interpret data on a pie chart 2. Represent data on pie charts	Pie charts, Mathematical set, Data chart on activities.	Students to represent data from his environment on a pie chart